Assignment 9: SQL Date & Time Functions

Overview:

Practice SQL date and time functions by calculating ages, experience, and extracting components. Learn to filter dates dynamically.

1. Create and Populate Table

```sql
CREATE TABLE Employees (
  EmpID INT PRIMARY KEY,
  Name VARCHAR(50),
  DateOfBirth DATE,
  JoinDate DATE
);
INSERT INTO Employees (EmpID, Name, DateOfBirth, JoinDate) VALUES
  (1, 'Alice Johnson', '1985-05-20', '2010-06-15'),
  (2, 'Bob Smith',       '1990-08-10', '2015-09-01'),
  (3, 'Charlie Brown', '1988-03-25', '2012-11-12'),
  (4, 'Diana Prince',  '1992-01-30', '2017-07-08'),
  (5, 'Eve Adams',      '1987-12-05', '2013-03-20');
```

Submission Type : Write the SQL query in the **PDF/Doc/Word** file and Take a screenshot of this part Output **(as screenshot_1)** and save it in the same **PDF/Doc/Word** file.

2. SQL Query Tasks

1. **a) Current Date & Time:** Display the current system date and time.
SQL:
SELECT CURRENT_TIMESTAMP;

Submission Type : *Write the SQL query in the **PDF/Doc/Word** file and Take a screenshot of this part Output **(as screenshot_2a)** and save it in the same **PDF/Doc/Word** file.*

2. **b) Calculate Age:** Compute each employee's age in years based on DateOfBirth.
SQL:
SELECT Name, FLOOR(DATEDIFF(CURRENT_DATE, DateOfBirth)/365.25) AS Age
FROM Employees;

Submission Type : *Write the SQL query in the **PDF/Doc/Word** file and Take a screenshot of this part Output **(as screenshot_2b)** and save it in the same **PDF/Doc/Word** file.*

3. **c) Calculate Experience:** Compute total years of service since JoinDate.
SQL:
SELECT Name, FLOOR(DATEDIFF(CURRENT_DATE, JoinDate)/365.25) AS YearsExperience FROM Employees;

Submission Type : *Write the SQL query in the* ***PDF/Doc/Word*** *file and Take a screenshot of this part Output* ***(as screenshot_2c)*** *and save it in the same* ***PDF/Doc/Word*** *file.*

4. **d) Extract DOB Components:** Extract year, month, and day from DateOfBirth.
SQL:
SELECT Name, YEAR(DateOfBirth) AS BirthYear, MONTH(DateOfBirth) AS BirthMonth, DAY(DateOfBirth) AS BirthDay FROM Employees;

Submission Type : *Write the SQL query in the* ***PDF/Doc/Word*** *file and Take a screenshot of this part Output* ***(as screenshot_2d)*** *and save it in the same* ***PDF/Doc/Word*** *file.*

5. **e) Born in August:** List employees born in the month of August.
SQL:
SELECT Name FROM Employees WHERE MONTH(DateOfBirth) = 8;

Submission Type : *Write the SQL query in the* ***PDF/Doc/Word*** *file and Take a screenshot of this part Output* ***(as screenshot_2e)*** *and save it in the same* ***PDF/Doc/Word*** *file.*

6. **f) Upcoming Birthdays:** Display names of employees whose birthdays fall in the next 30 days.
SQL:
SELECT Name FROM Employees WHERE
DATE_FORMAT(DATE_ADD(DateOfBirth, INTERVAL YEAR(CURRENT_DATE)-YEAR(DateOfBirth) YEAR), '%m-%d')
BETWEEN DATE_FORMAT(CURRENT_DATE, '%m-%d') AND DATE_FORMAT(DATE_ADD(CURRENT_DATE, INTERVAL 30 DAY), '%m-%d');

Submission Type : *Write the SQL query in the* ***PDF/Doc/Word*** *file and Take a screenshot of this part Output* ***(as screenshot_2f)*** *and save it in the same* ***PDF/Doc/Word*** *file.*

Submission Guidelines:

Create One Main Folder (named Assignment9_YourName) which includes -

- One **Doc/PDF/Word** file containing all the Queries and the screenshots of each part.
- Put your SQL file as **Assignment9_YourName.sql**

Convert the folder into ZIP format.
Upload the ZIP folder on the submission portal.

Watch this to learn how to create ZIP file

https://youtu.be/c7x-Q3AQEgw?si=_mS4c7OXADQ7DroE

If you have any doubt, you can directly contact the Tutedude mentor.